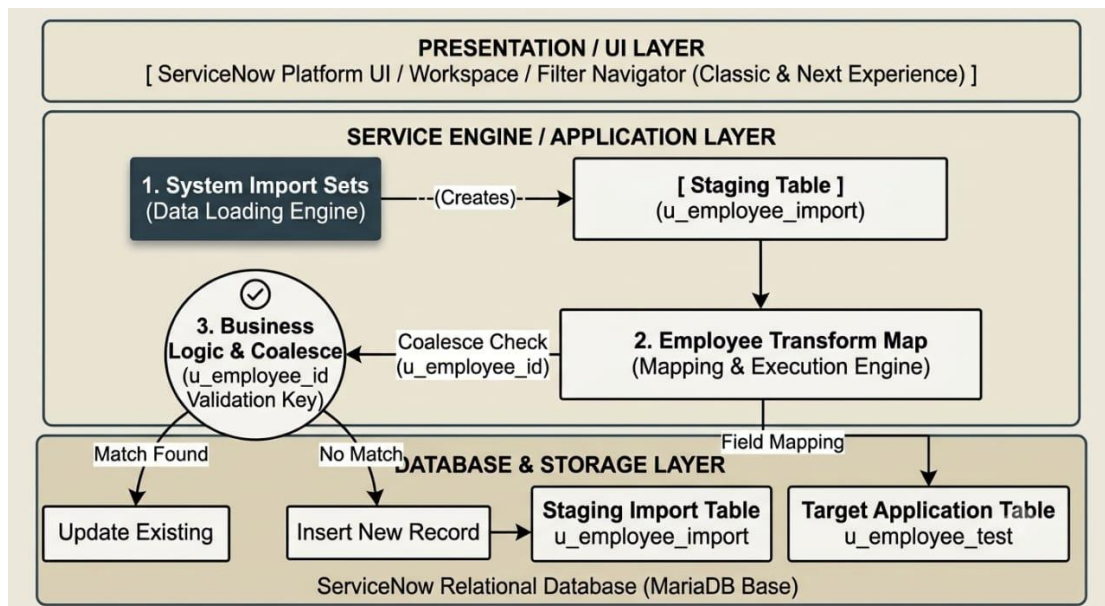


Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28 september 2026
Team ID	SWTID-2026-4908
Project Name	Import Data using Transform Maps
Maximum Marks	4 Marks

1. Technical Architecture (Text-Based Flow Diagram):



2.Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Platform administration interface used to upload files, map fields, and monitor import status	ServiceNow Polaris / Next Experience UI (HTML5, CSS3, JavaScript, Jelly).
2.	Application Logic-1	Staging Engine (System Import Sets) that processes uploaded files and constructs the temporary import table	ServiceNow Server-Side Glide API & Import Set Subsystem.
3.	Application Logic-2	Transformation Engine (Transform Map) executing auto-field matching and data casting.	ServiceNow Transform Engine / JavaScript (Rhino Engine).
4.	Application Logic-3	Coalesce Logic checking existing target rows against u_employee_id to perform insert vs. update branching.	ServiceNow Data Scrubbing & Coalesce Engine.
5.	Database	Multi-tenant relational schema housing custom	ServiceNow Instance Database Engine (MariaDB-

		tables (u_employee_import and u_employee_test).	backed relational store).
6.	Cloud Database	High-availability cloud-hosted database underlying the ServiceNow instance.	ServiceNow Advanced High Availability (AHA) Cloud Infrastructure.
7.	File Storage	Staging repository for raw .xlsx file attachments prior to parsing.	ServiceNow Attachment Table (sys_attachment) / Instance Cloud Storage
8.	External API-1	Ingestion interface allowing programmatic spreadsheet submission.	ServiceNow Table API / Import Set REST API.
9.	External API-2	Authentication layer ensuring secure administrative session access.	OAuth 2.0 / Multi-Factor Authentication (MFA).
10.	Machine Learning Model	N/A (Rule-based deterministic field mapping used instead).	Native Auto-Map Matching Algorithm
11.	Infrastructure	Multi-instance enterprise cloud architecture hosted	Cloud Infrastructure (ServiceNow

		on ServiceNow infrastructure	Personal Developer Instance - PDI / Hosted Cloud).
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2. Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Native JavaScript libraries utilized for data parsing and browser UI rendering	JavaScript (ECMAScript), Apache POI (underlying Excel parser engine).
2.	Security Implementations	Access restricted strictly to authorized roles (admin, import_admin) controlling staging table read/write permissions.	Role-Based Access Control (RBAC), Access Control Lists (ACLs), TLS 1.3 Encryption in transit
3.	Scalable Architecture	Enterprise 3-tier architecture isolating client browser, app server nodes, and dedicated database clusters.	ServiceNow Multi-Node Cloud Architecture (Application Nodes + DB Replica)
4.	Availability	Instance redundancy ensured across	ServiceNow Advanced High Availability (AHA)

		primary and secondary data center pairs.	Architecture with automated failover.
5.	Performance	Asynchronous batch transformation handling row insertions in memory prior to bulk committing	ServiceNow Import Set Batch Processing & In-Memory Transformation Engine.